

Serverless EC2 Scheduler (Start/Stop Instances using Lambda with Least Privilege)

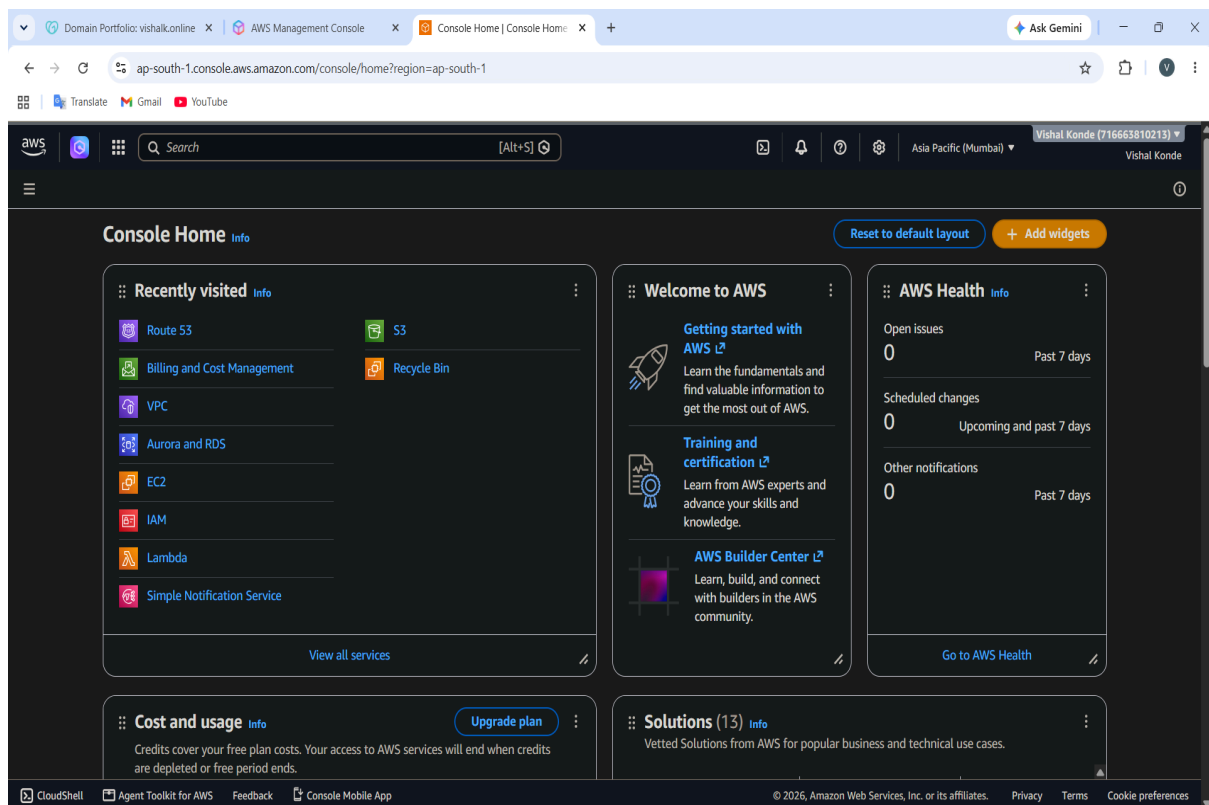
By : Vishal Konde

Project Goal:

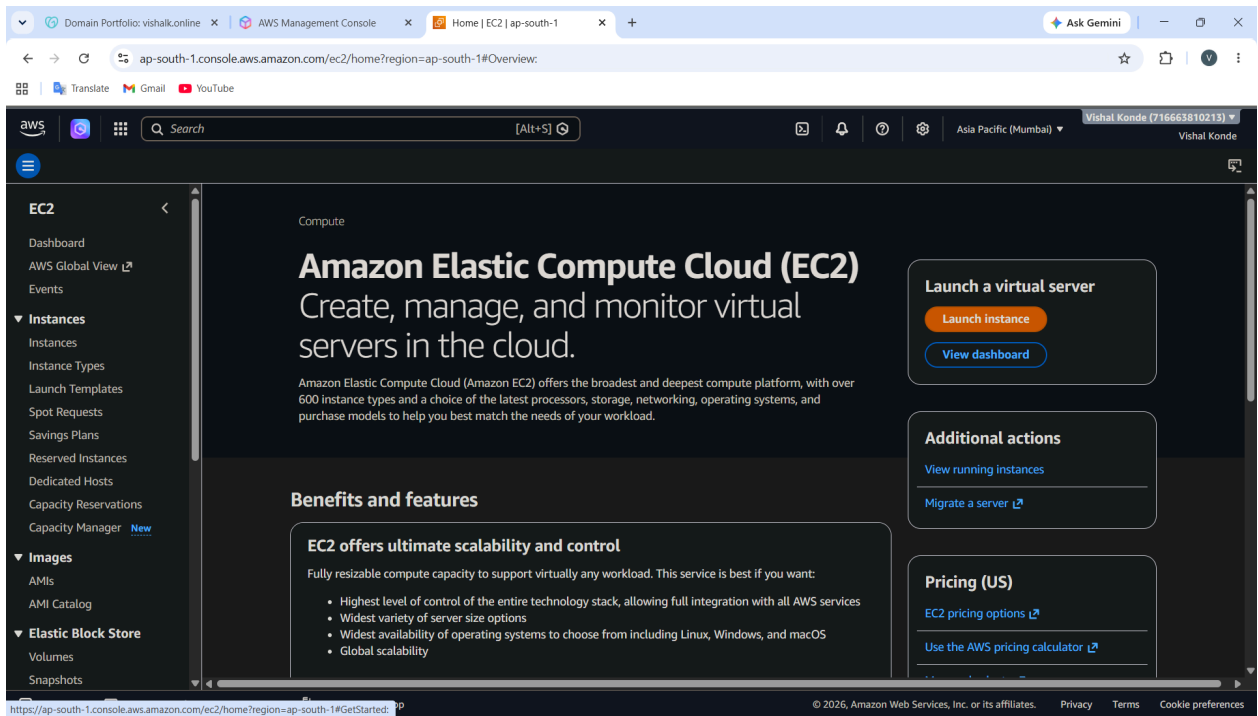
Automate EC2 start/stop using:

- AWS Lambda
- EventBridge (Scheduler)
- IAM (least privilege)
- Tags (dynamic targeting)

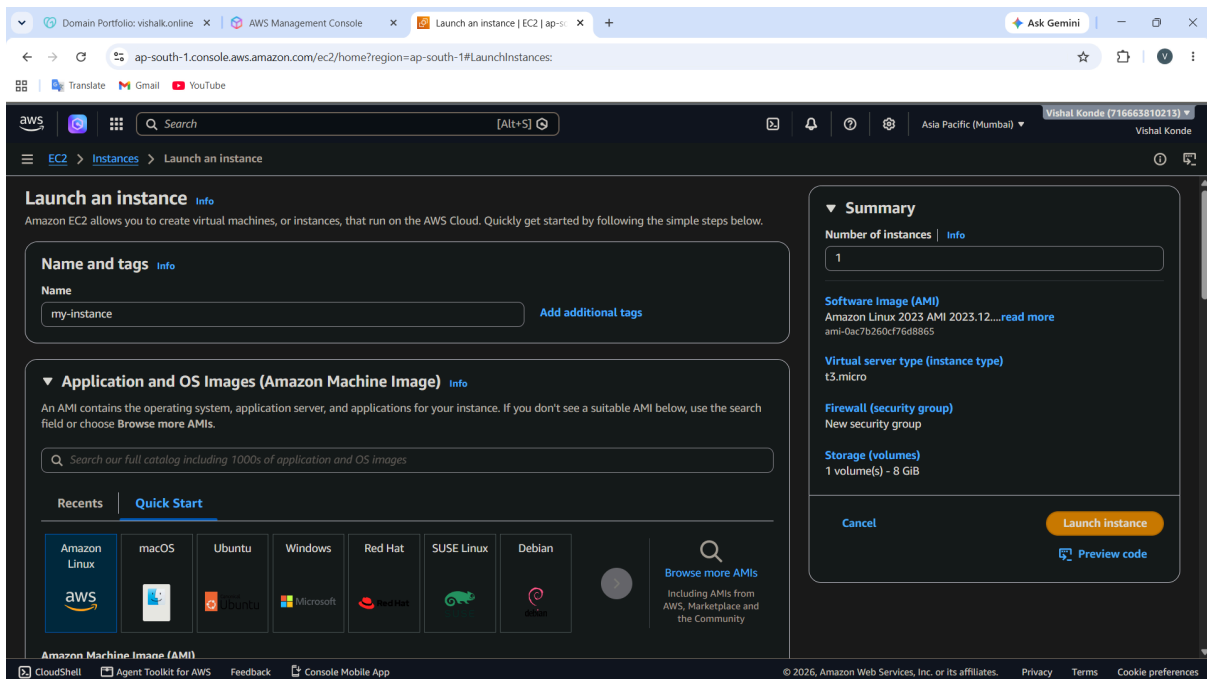
1) Login to AWS console



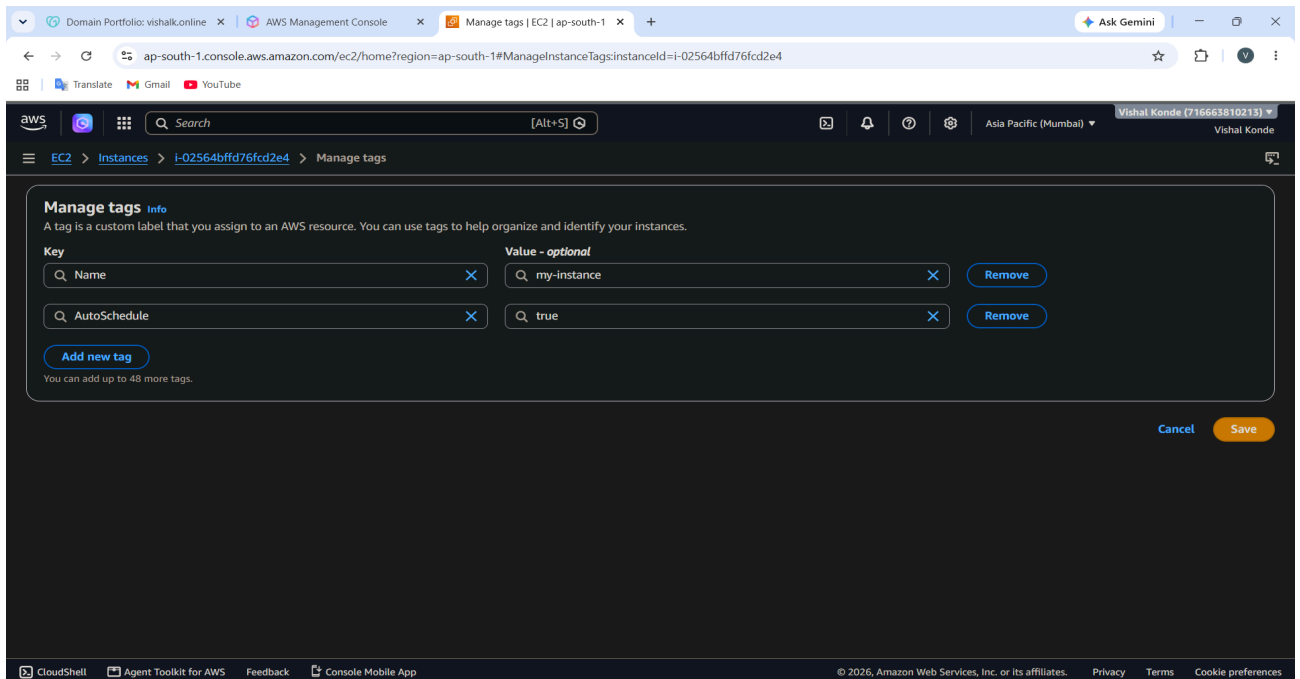
2) Come to EC2 service home page .



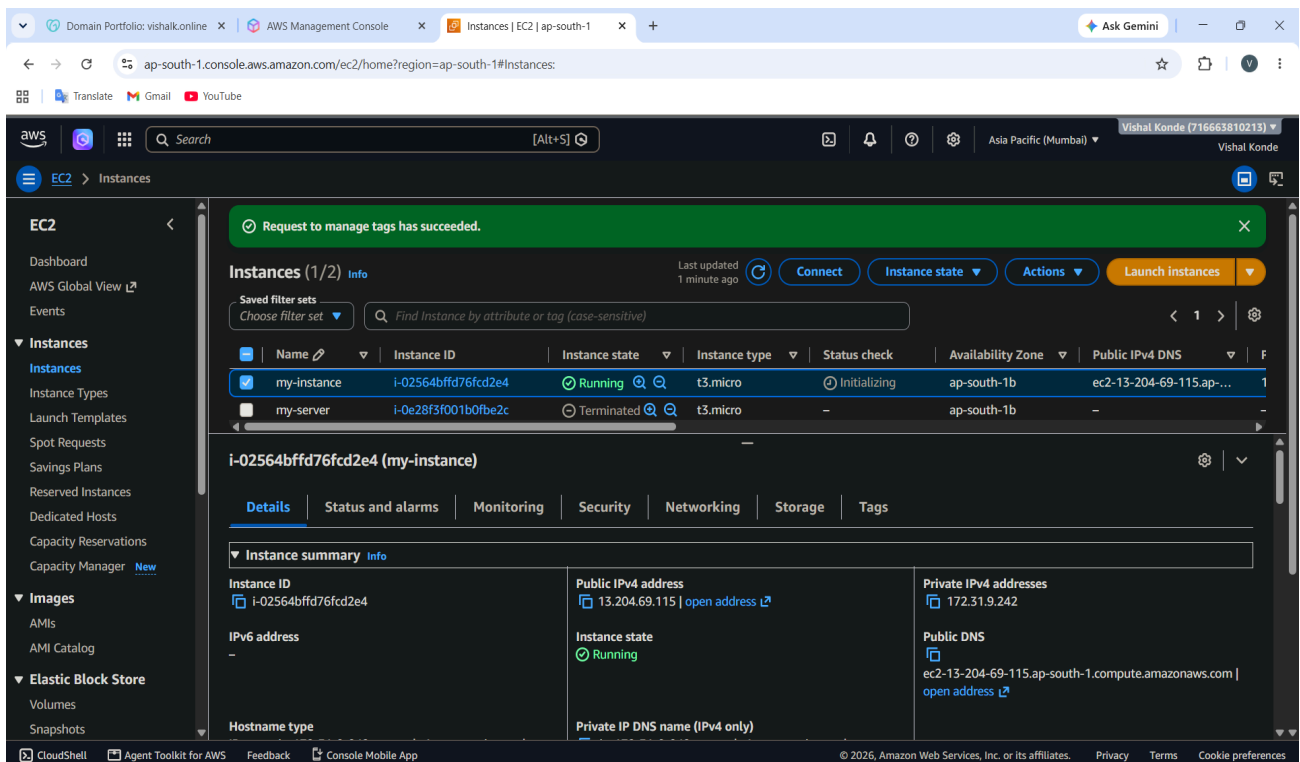
3) Launching an Instance .



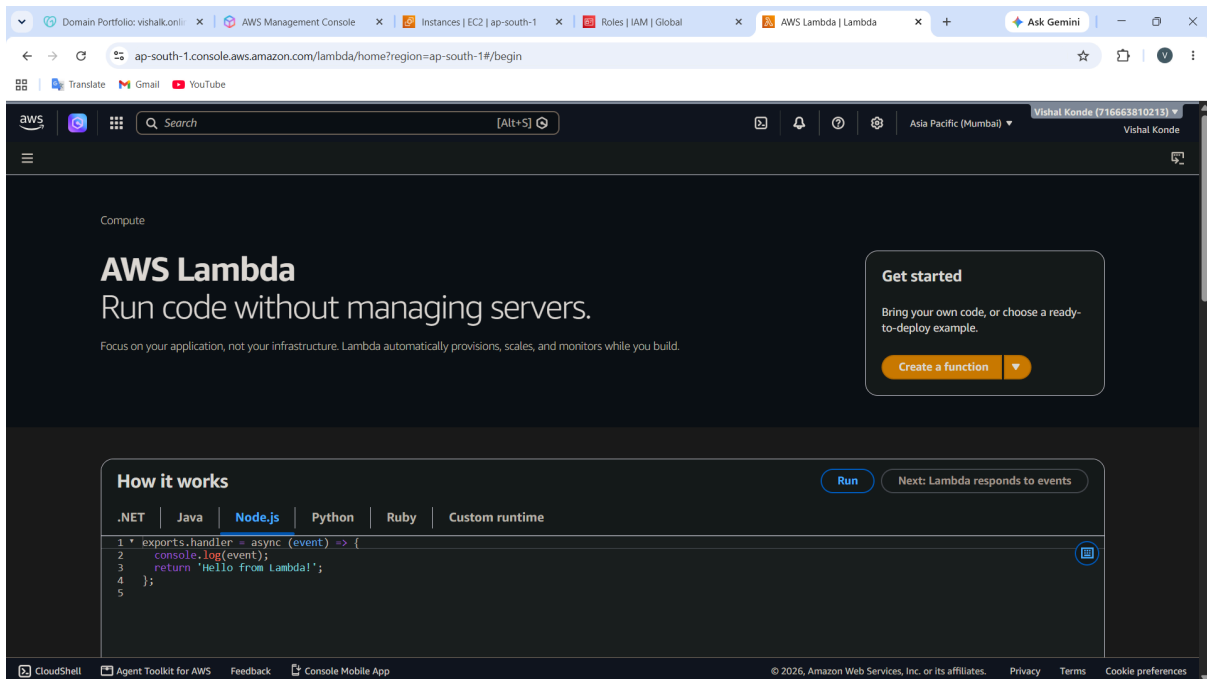
4) Tag Added to the Instance that prevents hardcoding instance id.



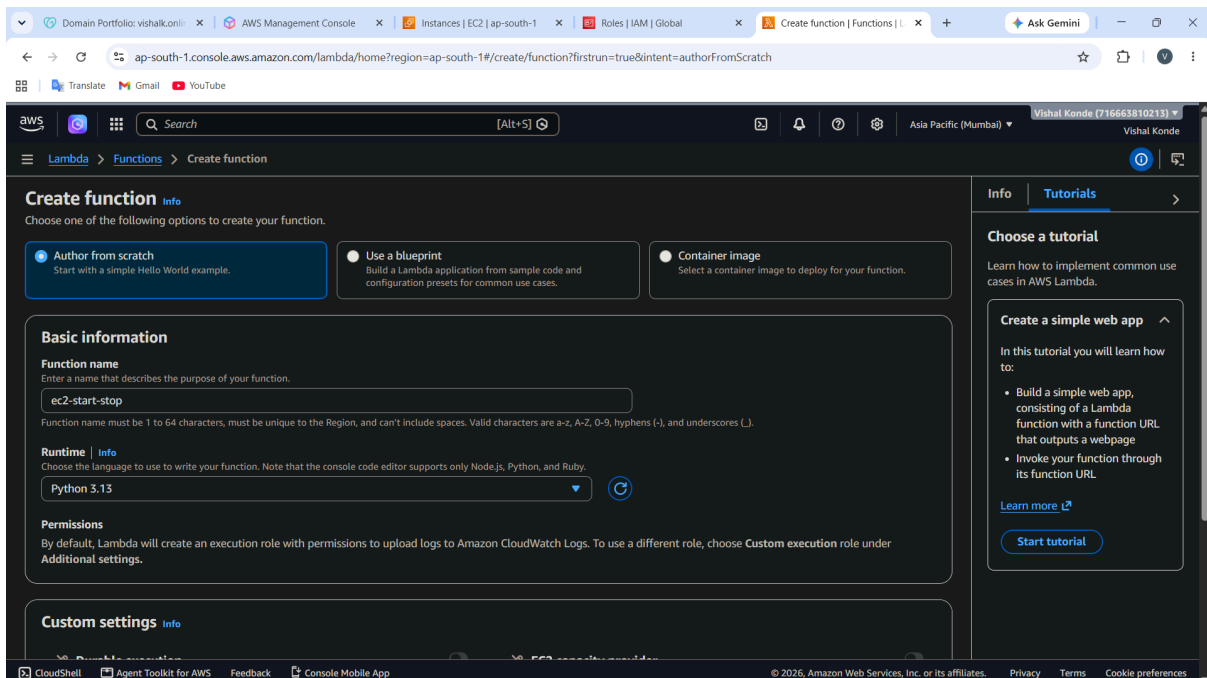
5) Instance Created Successfully.



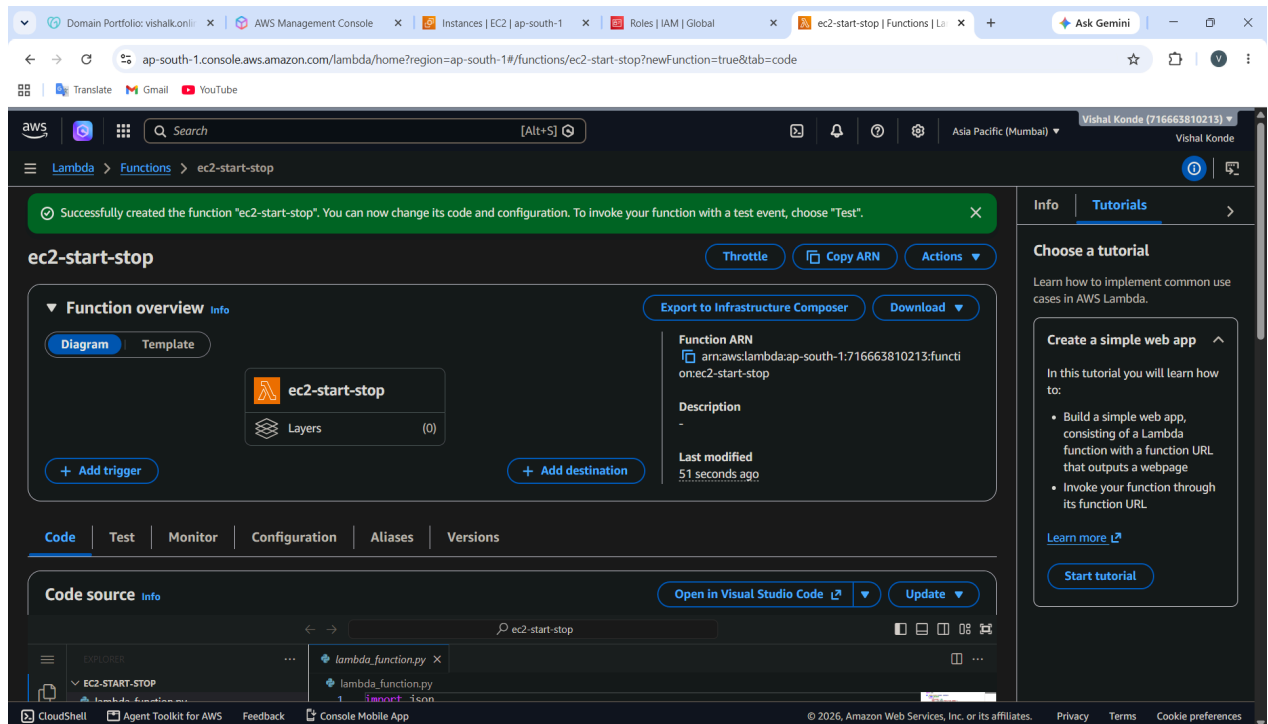
6) For Creating Lambda Function come to Lambda service home page.



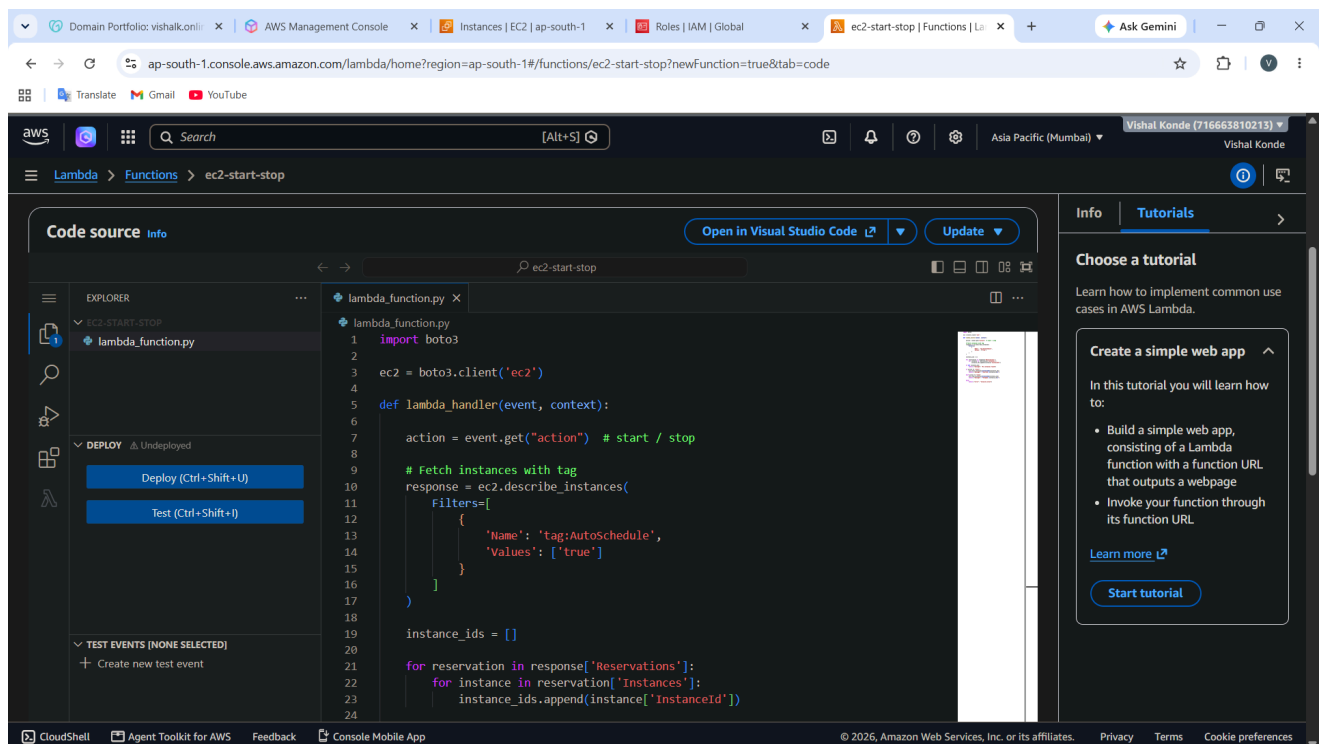
7) Creating Lambda function(Runtime used Python 3.13).



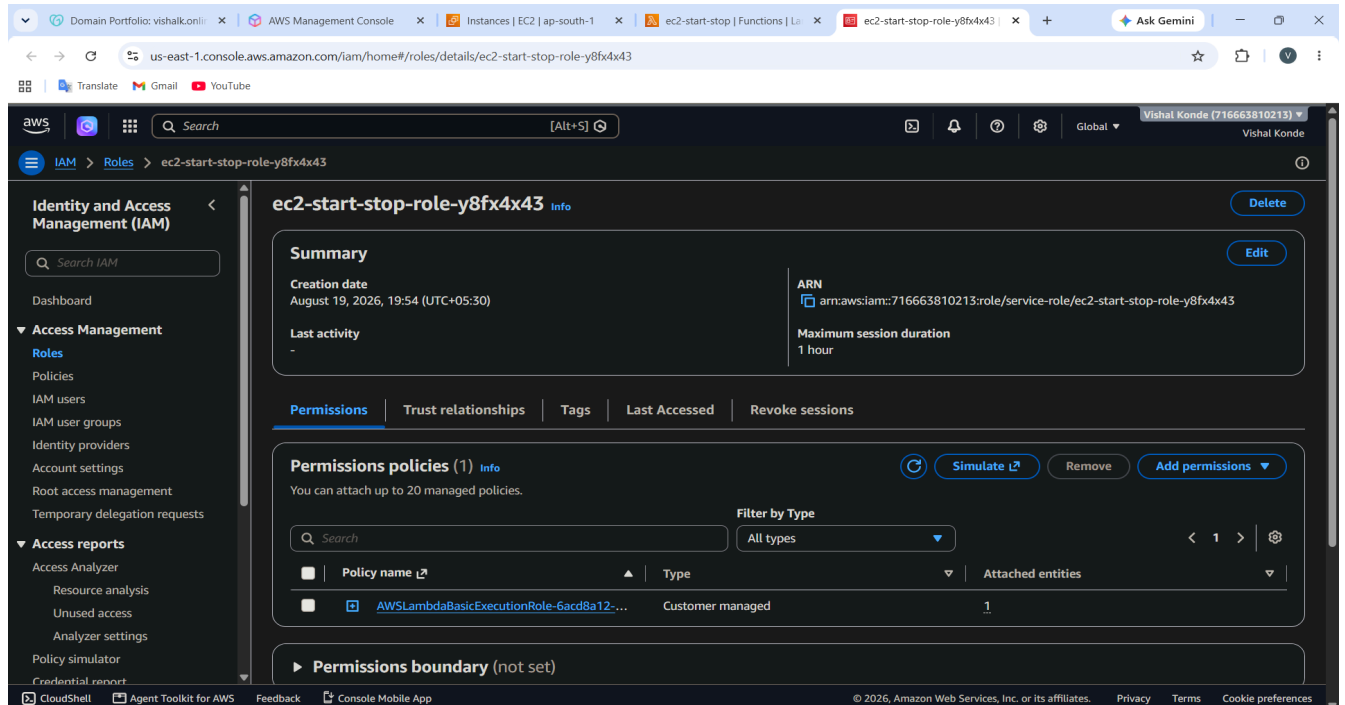
8) Lambda Function Created.



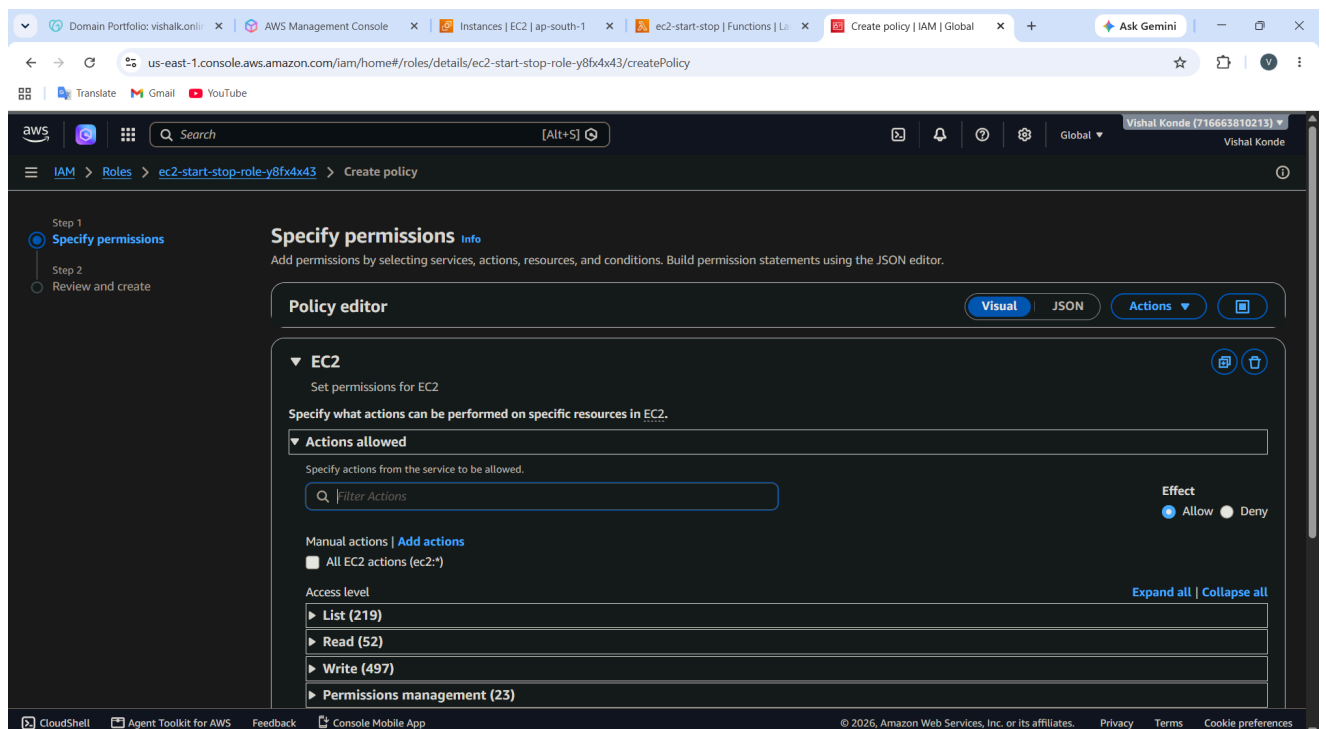
9) In Code section a VSstudio like Interface gets opened , Add code here and Deploy.

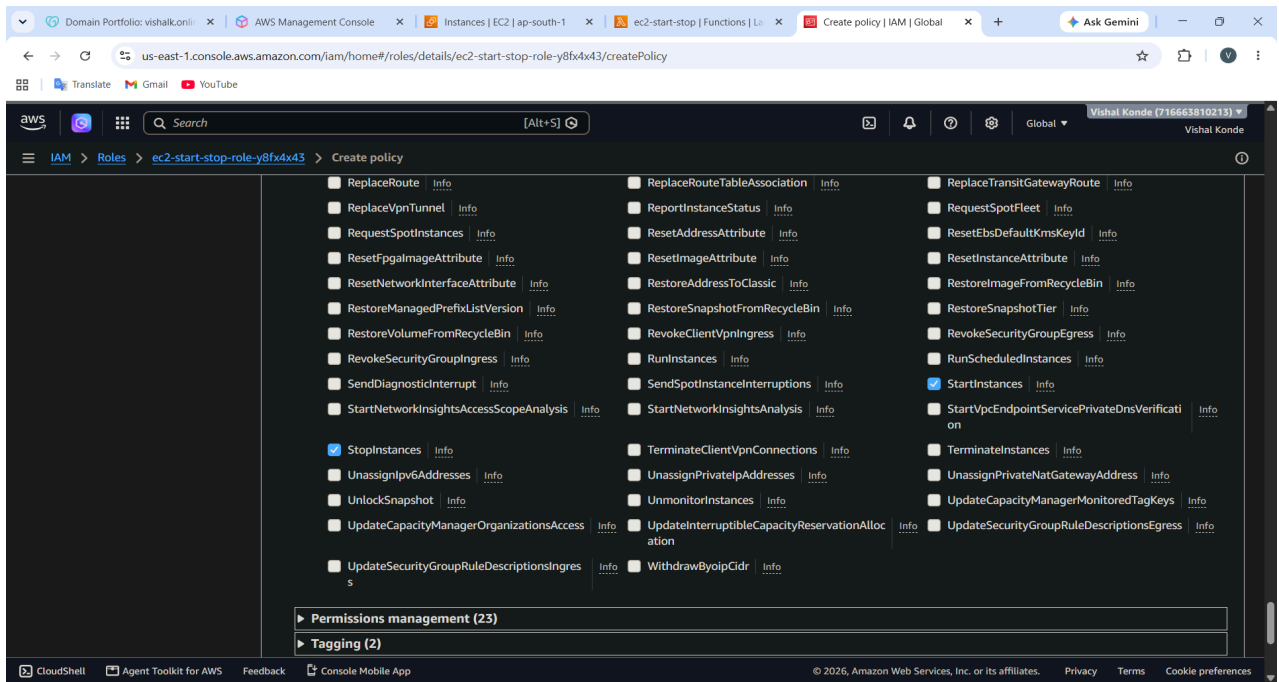


10) Role gets created for Lambda automatically. Add Policy/Permissions of EC2 to the Role.

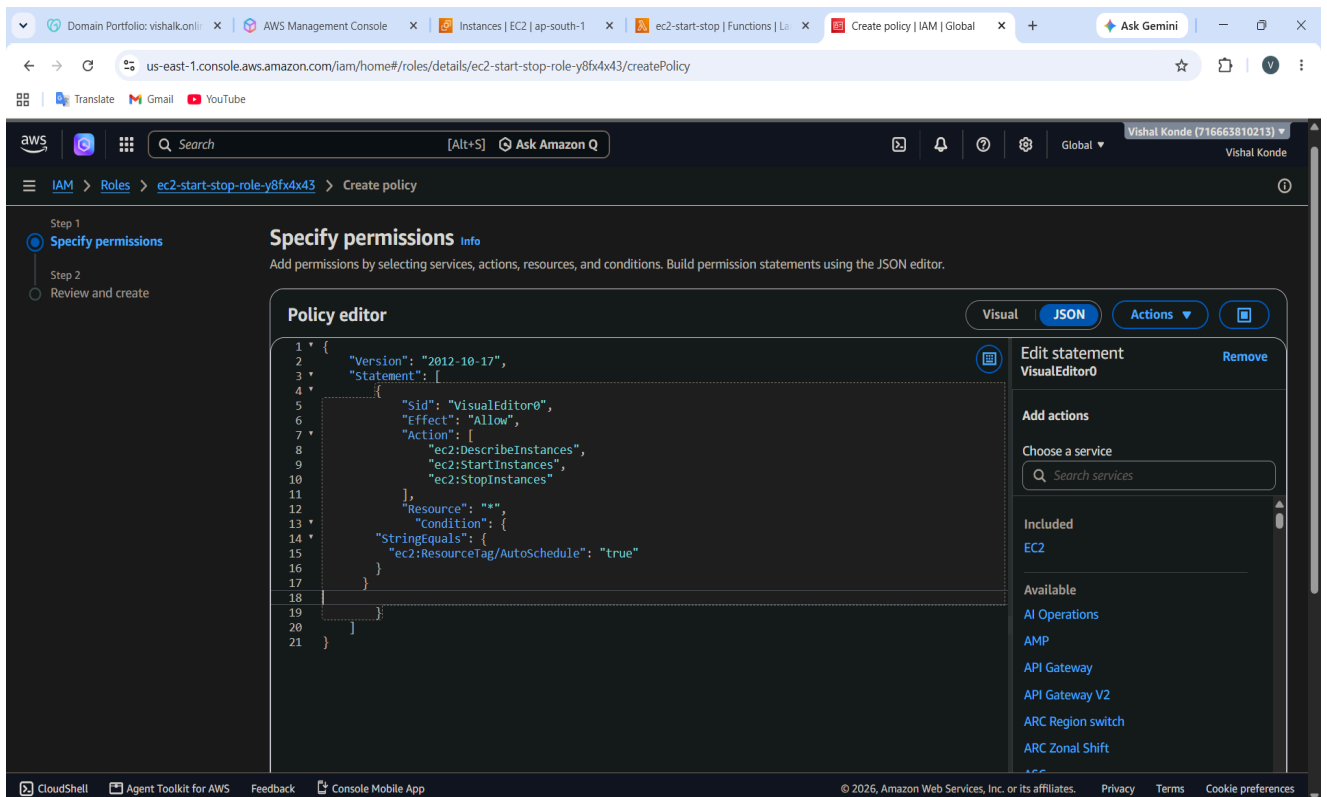


11) Create inline policy , Specify Permissions Of EC2.

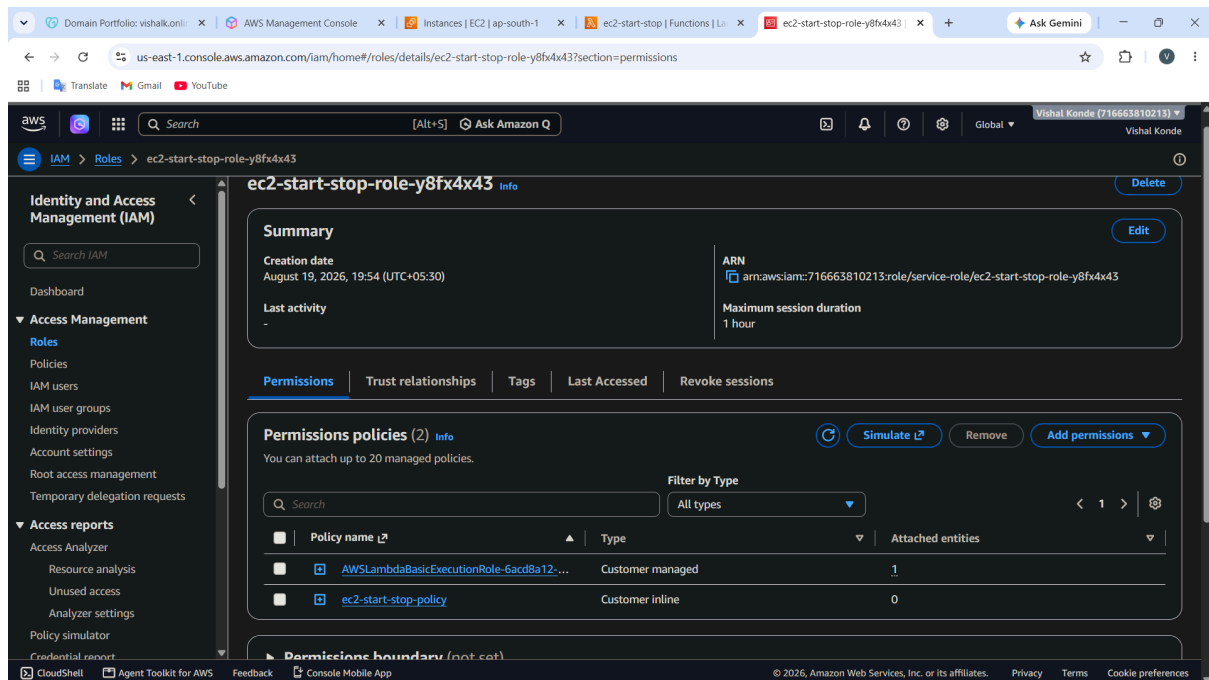




IAM Role with Condition(ResourceTag) , which helps to Identify the Instance

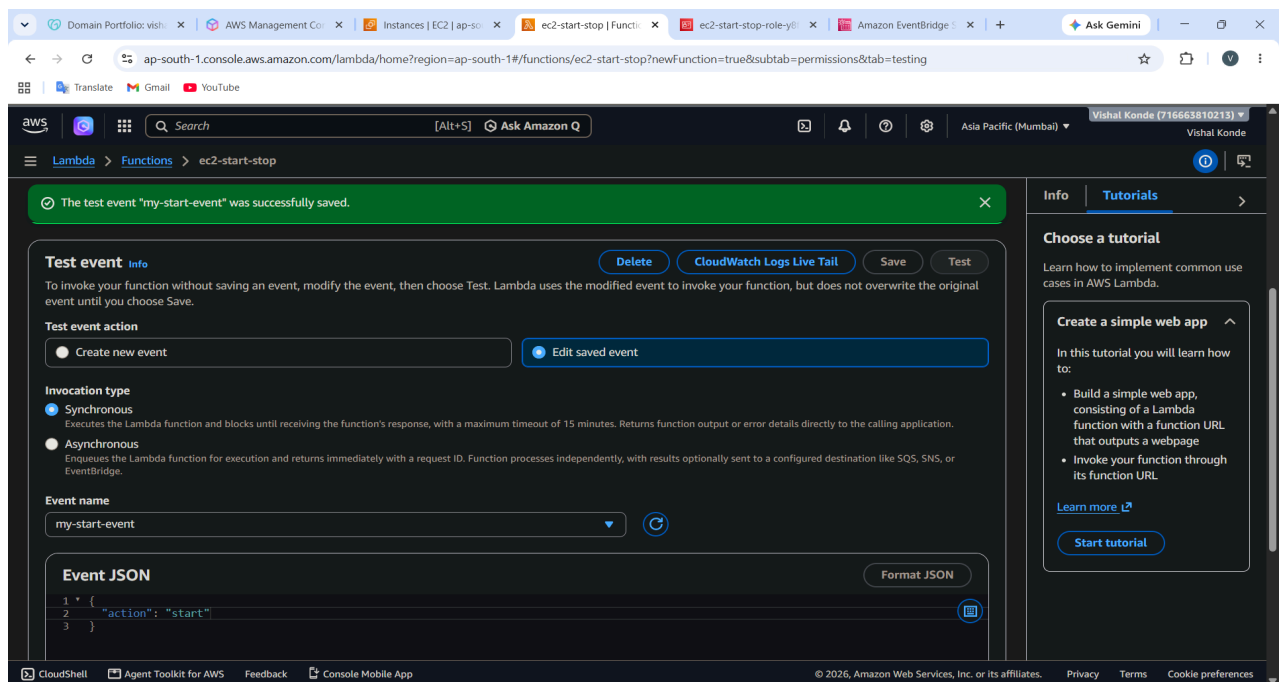


Policy Added to Lamda Role.

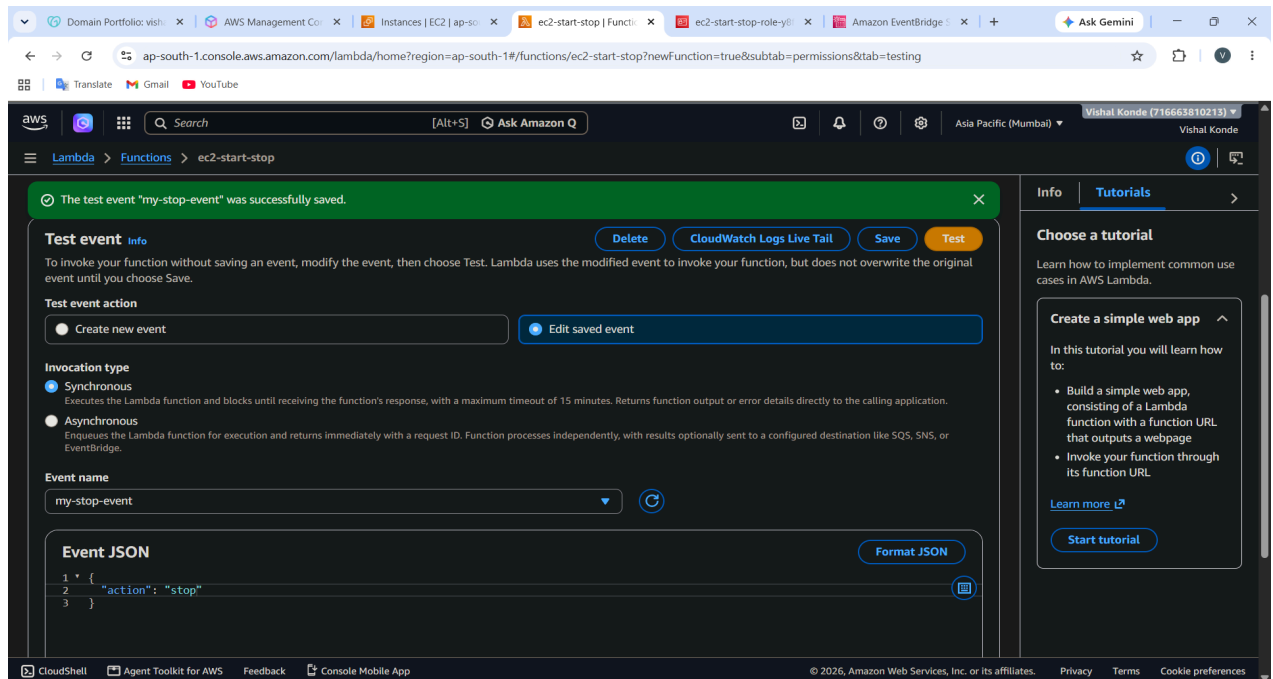


12) Create Tests for Lambda function.

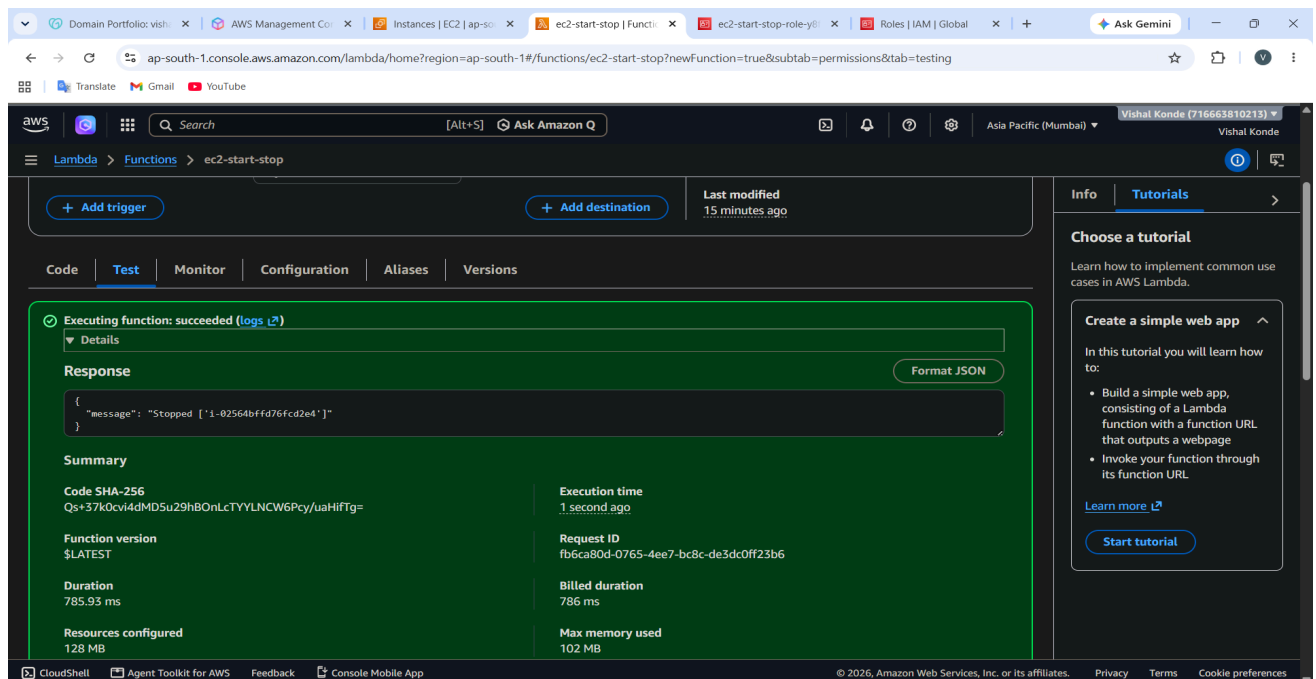
1) For Start: {"action": "start"}



11) For Stop : {"action": "stop"}



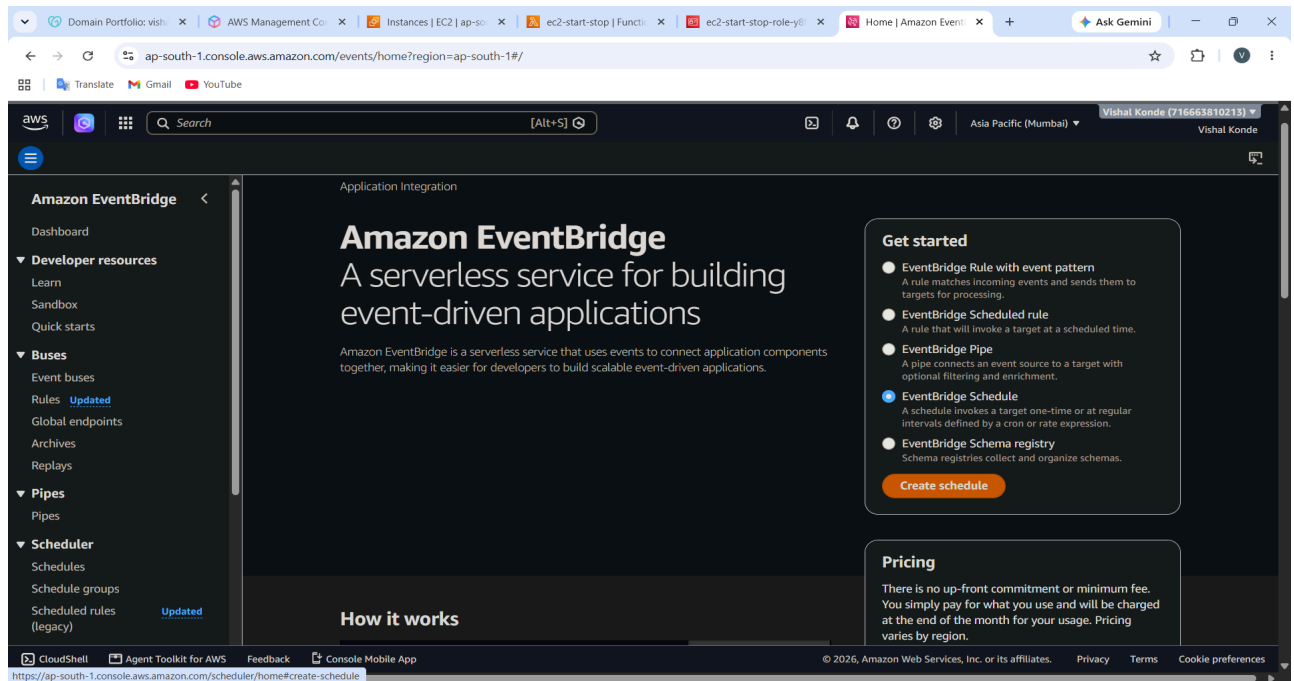
13) Save and Test the Lambda function is Working or not.



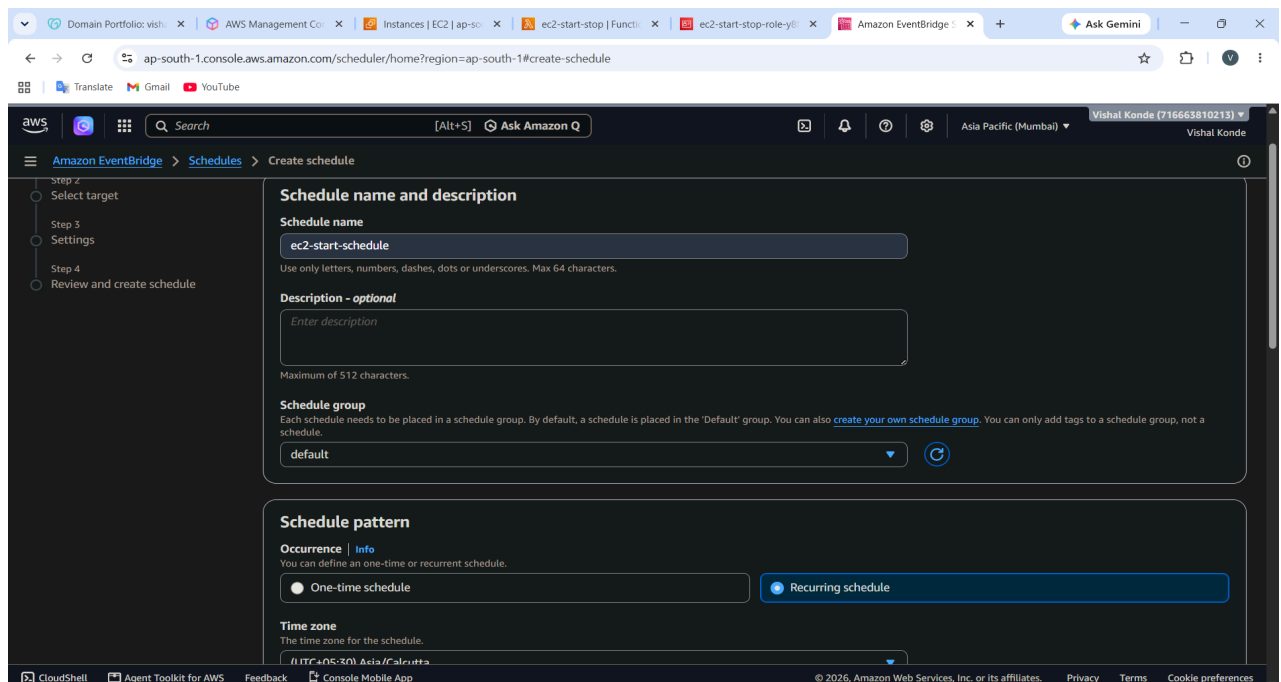
Execution is successful , Lambda is changing Ec2 Instance State(Start/Stop).

Now Creating Schedule using EventBridge service which invoke Lambda.

14) Go to EventBridge service home page and Create Schedules.



15) Create Schedule for Start.



16) Schedule Pattern = Recuuring Schedule with Cron Expression that invokes the Lambda Everyday At 6:00 AM .

The screenshot shows the AWS EventBridge console's 'Create schedule' page. The 'Occurrence' section has 'Recurring schedule' selected. The 'Time zone' is set to '(UTC+05:30) Asia/Calcutta'. Under 'Schedule type', 'Cron-based schedule' is selected. The 'Cron expression' field shows '0 6 * * ? *', with labels for Minutes, Hours, Day of month, Month, Day of week, and Year. Below this, 'Next 10 trigger dates' are listed, starting with 'Thu, 20 Aug 2026 06:00:00 (UTC+05:30)'. The page footer includes links for CloudShell, Agent Toolkit for AWS, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services.

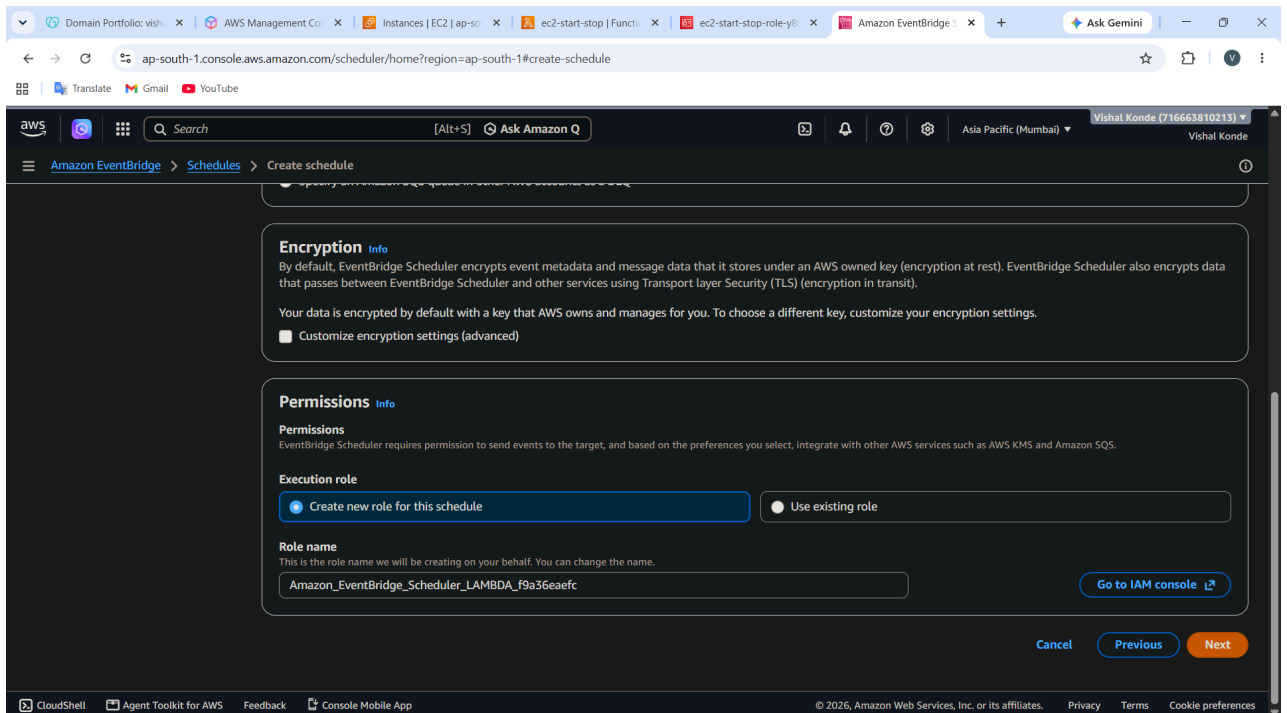
17) Select Target as Lambda And Payload of start.

The screenshot shows the 'Create schedule' page with the 'Invoke' target selected. The 'Invoke' section shows 'AWS Lambda' as the target type. The 'Lambda function' dropdown is set to 'ec2-start-stop'. The 'Payload' section shows a JSON configuration:

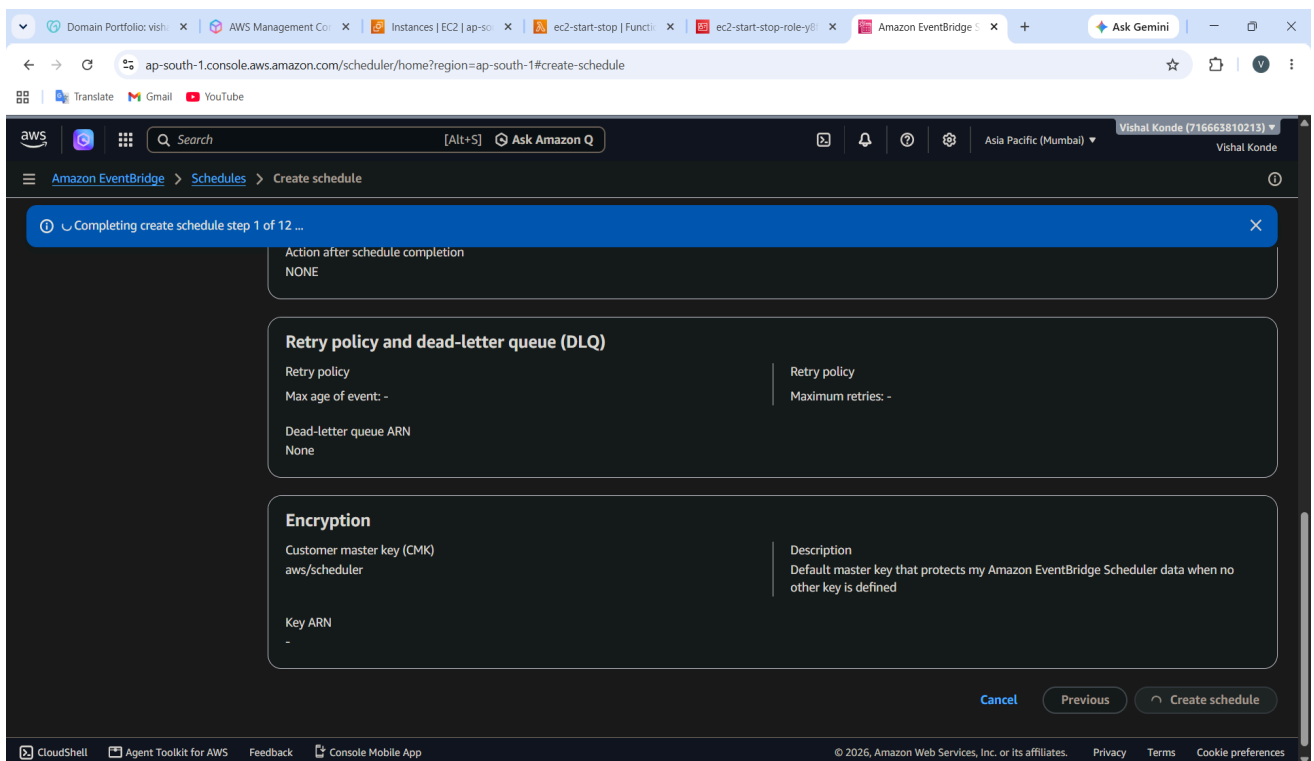
```
{  "action": "start"}
```

. The page footer includes links for CloudShell, Agent Toolkit for AWS, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services.

18) In Permission select create new Role(Automatically creates role(EventBridge) with Permissions of Lambda).



19) Review And Create Schedule.



20) Create Schedule for Stop.

The screenshot shows the AWS EventBridge console in the 'Specify schedule detail' step. The left sidebar indicates the steps: Step 1 (Specify schedule detail), Step 2 (Select target), Step 3 (Settings), and Step 4 (Review and create schedule). The main content area is titled 'Specify schedule detail' and contains two sections: 'Schedule name and description' and 'Schedule pattern'.

Schedule name and description

- Schedule name:** A text input field containing 'ec2-stop-schedule'. Below it, a note states: 'Use only letters, numbers, dashes, dots or underscores. Max 64 characters.'
- Description - optional:** A text input field with the placeholder 'Enter description'. Below it, a note states: 'Maximum of 512 characters.'
- Schedule group:** A dropdown menu with 'default' selected. A note states: 'Each schedule needs to be placed in a schedule group. By default, a schedule is placed in the "Default" group. You can also [create your own schedule group](#). You can only add tags to a schedule group, not a schedule.'

Schedule pattern

Occurrence | Info
You can define an one-time or recurrent schedule.

☐ One-time schedule ☒ Recurring schedule

21) Recurring Schedule Everynight At 12:00 AM.

The screenshot shows the AWS EventBridge console in the 'Schedule pattern' step. The left sidebar indicates the steps: Step 1 (Specify schedule detail), Step 2 (Select target), Step 3 (Settings), and Step 4 (Review and create schedule). The main content area is titled 'Schedule pattern' and contains three sections: 'Occurrence', 'Time zone', and 'Schedule type'.

Schedule pattern

Occurrence | Info
You can define an one-time or recurrent schedule.

☐ One-time schedule ☒ Recurring schedule

Time zone
The time zone for the schedule.

(UTC+05:30) Asia/Calcutta

Schedule type
Choose the schedule type that best meets your needs.

☒ Cron-based schedule
A schedule set using a cron expression that runs at a specific time, such as 8:00 a.m. PST on the first Monday of every month.

☐ Rate-based schedule
A schedule that runs at a regular rate, such as every 10 minutes.

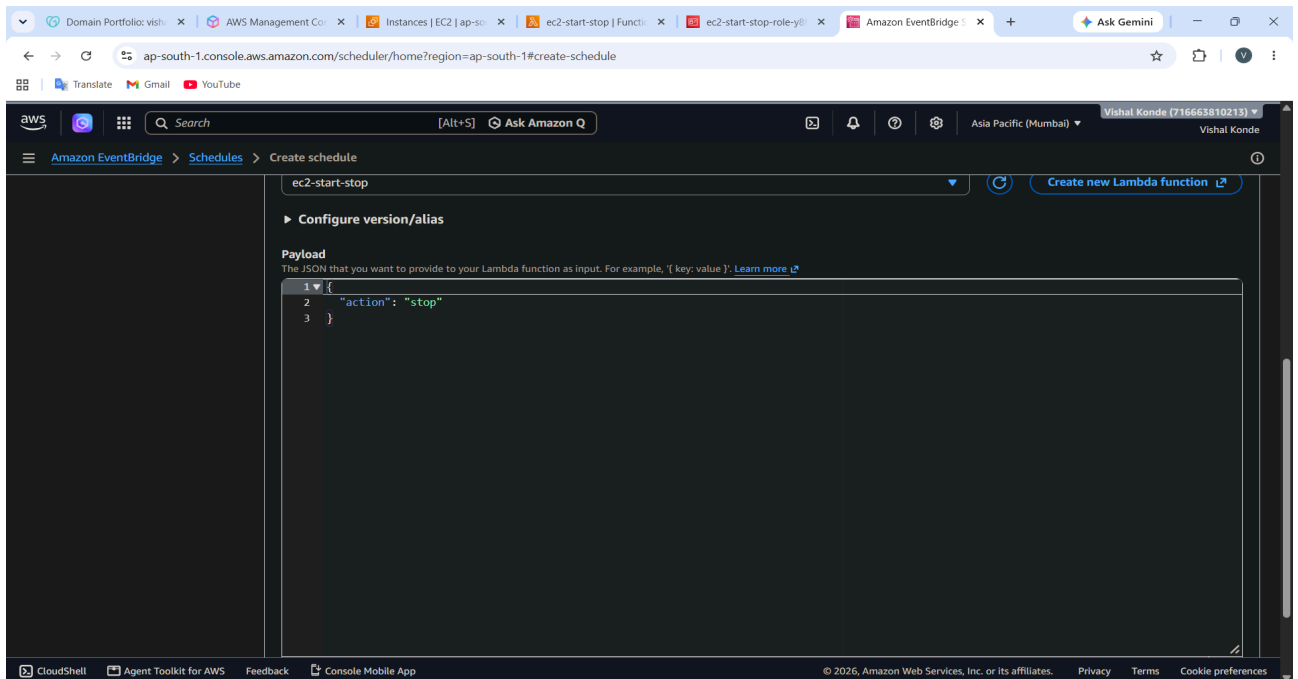
Cron expression | Info
Define the cron expression for the schedule

cron (0 Minutes 0 Hours * Day of month * Month ? Day of the week * Year)

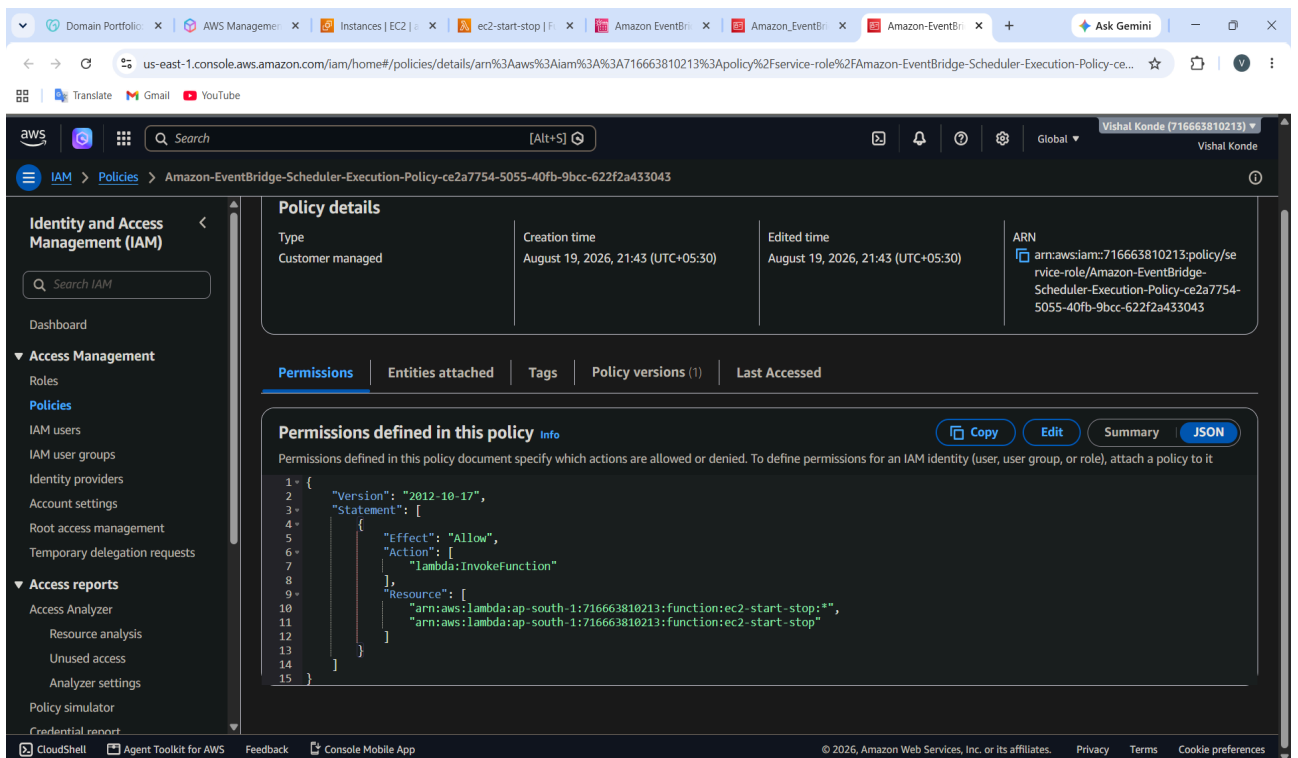
Next 10 trigger dates
Date and time are displayed in your current time zone in UTC format, e.g. "Wed, Nov 9, 2022 09:00 (UTC - 08:00)" for Pacific time

Thu, 20 Aug 2026 00:00:00 (UTC+05:30)
Fri, 21 Aug 2026 00:00:00 (UTC+05:30)

22) Target Invoke Lambda , Select Lambda Function and Stop Payload.



23) Role is Created Automatically for EventBridge Schedules with the permissions of Lambda (Invoke Function).



✓ As per the Schedule Time Lambda function gets invoke and Automatically Start/stop the Instance Successfully.

Start at 6:00 AM Daily

Stop at 12:00 AM Daily

The screenshot shows the AWS Management Console for the 'ap-south-1' region. The main content area displays the 'Instances (1)' page. A table lists the instances, with one instance in a 'Stopping' state. Below the table, there are three summary cards: 'Instance alarm states' (None), 'Service health' (Operating normally), and 'Health events affecting EC2 instances' (None). The left sidebar shows the navigation menu with categories like EC2, Images, and Elastic Block Store.

Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Ela
i-02564bffd76fcd2e4	Stopping	t3.micro	-	ap-south-1b	ec2-13-127-18-110.ap-...	13.127.18.110	-

Instance alarm states
In alarm
None Show instances in alarm

Service health
EC2 service status
Operating normally

Health events affecting EC2 instances
Open events
None Show instances
Upcoming events
None Show instances

Conclusion : This project automates EC2 instance start/stop operations using AWS Lambda and EventBridge. IAM least-privilege policies ensure the scheduler has only the required permissions. EC2 tags enable dynamic and flexible selection of instances without hardcoding IDs.

Overall, it reduces AWS costs, improves resource management, and provides a production-ready automation approach.

- **AWS Services Used**

EC2

Lambda

EventBridge

IAM